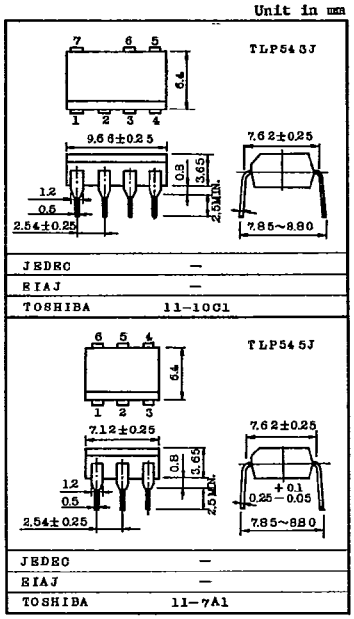
TLP543J, TLP545J

GaAs IRED & PHOTO-THYRISTOR

The TOSHIBA TLP543J consists of a phot-thyristor optically coupled to a gallium arsenide infrared emitting diode in a seven lead plastic DIP package.
The TOSHIBA TLP545J consists of a photo-thyristor optically coupled to a gallium arsenide infrared emitting diode in a six lead plastic DIP package.

- Peak Off-State Voltage : 600V Min.
- Trigger LED Current : 10mA Max.
- On-State Current : 150mA Max.
- Isolation Voltage : 2500Vrms Min.
- UL Recognized : File No. E67349

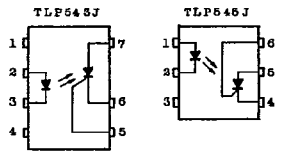
Unit in mm



MAXIMUM RATINGS (Ta = 25°C)

	CHARACTERISTIC	SYMBOL	RATING	UNIT
LED	Forward Current	I_F	60	mA
	Forward Current Derating (Ta≥39°C)	ΔI_F/°C	-0.7	mA/°C
	Peak Forward Current (100μs pulse, 100pps)	I_{FP}	1	A
	Power Dissipation	P_D	100	mW
	Power Dissipation Derating (Ta≥25°C)	ΔP_D/°C	-1.0	mW/°C
	Reverse Voltage	V_R	5	V
	Junction Temperature	T_j	125	°C
DETECTOR	Peak Forward Voltage (R_{GK}=27KΩ)	V_{DRM}	600	V
	Peak Reverse Voltage (R_{GK}=27KΩ)	V_{RRM}	600	V
	On-State Current	$I_{T(RMS)}$	150	mA
	On-State Current Derating (Ta≥25°C)	ΔI_T/°C	-2.0	mA/°C
	Peak On-State Current (100μs pulse, 120pps)	I_{TP}	3	A
	Peak One Cycle Surge Current	I_{TSM}	2	A
	Peak Reverse Gate Voltage	V_{GM}	5	V
	Power Dissipation	P_D	150	mW
	Power Dissipation Derating (Ta≥25°C)	ΔP_D/°C	-2.0	mW/°C
	Junction Temperature	T_j	100	°C
Storgat Temperature Range		Tstg	-55~150	°C
Operating Temperature Range		Topr	-55~100	°C
Lead Soldering Temperature (10sec.)		Tsold	260	°C
Total Package Power Dissipation		P_T	250	mW
Total Package Power Dissipation Derating (Ta≥25°C)		ΔP_T/°C	-3.3	mW/°C
Isolation Voltage (AC, 1 min, RH≤60%)		BVS	2500	Vrms

PIN CONFIGURATIONS (TOP VIEW)



TLP543J
1: NC
2: ANODE
3: CATHODE
4: NC
5: GATE
6: CATHODE
7: ANODE

TLP545J
1: ANODE
2: CATHODE
3: NC
4: CATHODE
5: ANODE
6: GATE

RECOMMENDED OPERATING CONDITIONS

CHARACTERISTIC	SYMBOL	MIN.	TYP.	MAX.	UNIT
Supply Voltage	V_{AC}	-	-	240	Vac
Forwar Current	I_F	15	20	25	mA
Operating Temperature	Topr	-25	-	85	°C
Gate to Cathode Resistance	R_{GK}	-	10	27	kΩ
Gate to Cathode Capacity	C_{GK}	-	0.01	0.1	μF

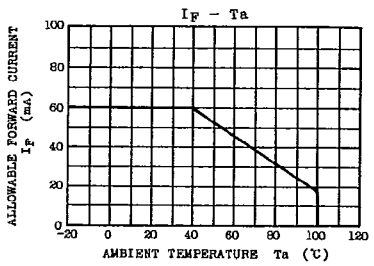
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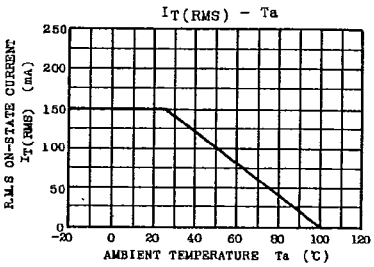
INDIVIDUAL ELECTRICAL CHARACTERISTICS (Ta=25°C)

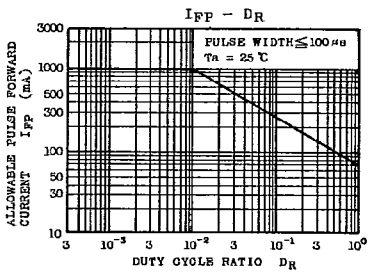
	CHARACTERISTIC	SYMBOL	TEST CONDITION		MIN.	TYP.	MAX.	UNIT
LED	Forward Voltage	V_F	I_F=10mA		1.0	1.15	1.3	V
	Reverse Current	I_R	V_R=5V		-	-	10	μA
	Capacitance	C_T	V=0, f=1MHz		-	30	-	pF
DETECTOR	Off-State Current	I_{DRM}	V_{AK}=600V, R_{GK}=27kΩ	Ta=25°C	-	10	5000	nA
				Ta=85°C	-	1	150	μA
	Reverse Current	I_{RRM}	V_{KA}=600V, R_{GK}=27kΩ	Ta=25°C	-	10	5000	nA
				Ta=85°C	-	1	150	μA
	On-State Voltage	V_{TM}	I_{TM}=100mA		-	0.9	1.3	V
	Holding Current	I_H	R_{GK}=27kΩ		-	0.2	-	mA
	Off-State dv/dt	dv/dt	V_{AK}=420V, R_{GK}=27kΩ		-	10	-	V/μs
	Capacitance	C_j	V=0, f=1MHz Anode to Gate Gate to Cathode		- -	20 350	- -	pF

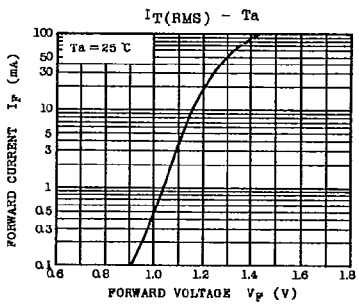
COUPLED CHARACTERISTICS (Ta=25°C)

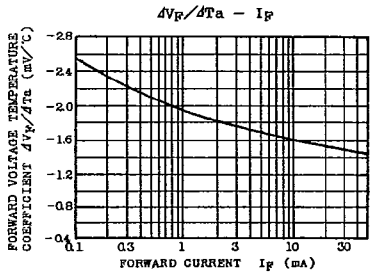
CHARACTERISTIC	SYMBOL	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
Trigger LED Current	I_{FT}	V_{AK}=6V, R_{GK}=27kΩ	-	5	10	mA
Turn-on Time	t_{on}	I_F=30mA, V_{AA}=50V, R_{GK}=27kΩ	-	10	-	μs
Coupled dv/dt	dv/dt	VS=500V, R_{GK}=27kΩ	500	-	-	V/μs
Capacitance Input to Output	C_S	V_S=0, f=1MHz	-	0.8	-	pF
Isolation Resistance	R_S	V_S=500V	5×10^{10}	10^{14}	-	Ω
Isolation Voltage	BV_S	AC, 1 minute	2500	-	-	V_{rms}
		AC, 1 second	-	5000	-	V_{rms}
		DC, 1 minute	-	5000	-	Vdc

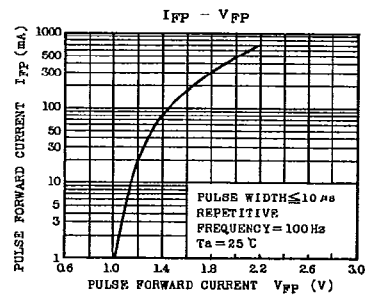






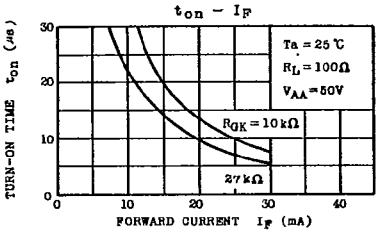




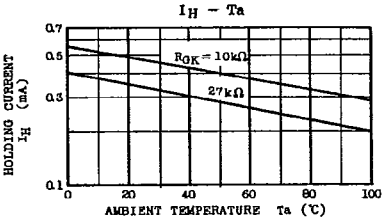


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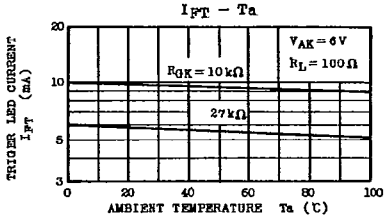
t_{on} – I_F



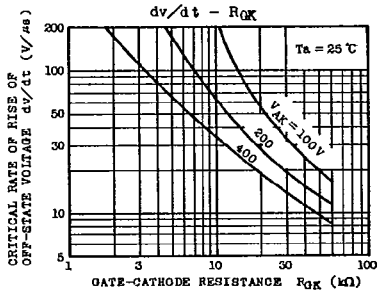
I_H – Ta



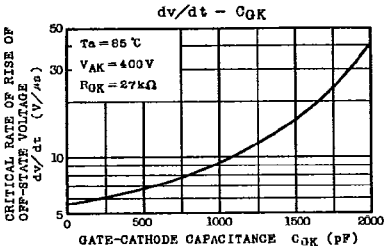
I_{FT} – Ta



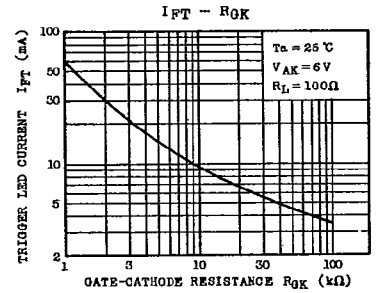
dv/dt – R_{GK}



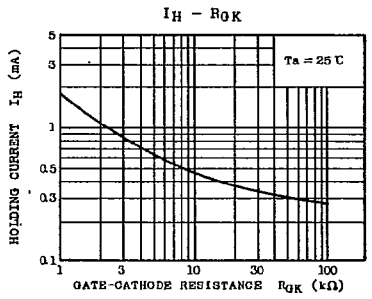
dv/dt – C_{GK}



I_{FT} – R_{GK}



I_H – R_{GK}



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